# Byla Sanjith

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 8.32 |
| Aditya Junior College Palakollu West Godavari District | Class XII (Board of Intermediate Education, Andhra Pradesh, India) | 2024 | 97.20% |
| Sri Chaitanya Techno(EM) School Palakollu West Godavari District | Class X (Board of Secondary Education, Andhra Pradesh, India) | 2022 | 97.33% |

## EXPERIENCE

**Texas Instruments | BYTE Program (Certified)**

- Successfully completed Texas Instruments' BYTE Program, strengthening fundamentals in analog circuit design through instructor-led training, technical assignments, and a proctored final evaluation.

## SKILLS

**Tools:** LTSpice | KiCad | Vivado | Docker | Git

**Programming:** C | Python | Verilog | Bash

## RELEVANT COURSES

**Analog:** Circuits and Network Analysis | Semiconductor Device Fundamentals | Analog Electronics | Control Theory for Circuits | Numerical Methods for Device Modeling

**Digital:** Microprocessor and Computer Architecture | Digital Systems | Productivity Tools for IC Design

**Others:** Signals and Systems | Introduction to CMOS Processing | Engineering Electromagnetics | Modern Physics

## PROJECTS

**Analog Circuit Design in LTspice**

- Built and verified OP27-based op-amp circuits, including practical integrator, differentiator, and Schmitt trigger topologies.
- Designed and simulated waveform generators (10 kHz square, triangle, and sawtooth) and oscillators (50 kHz RC phase-shift and Wien bridge).
- Simulated and analyzed 555 timer ICs in astable, monostable, and bistable configurations.
- Designed and simulated an LDO (low-dropout) voltage regulator using NMOS/PMOS current sources, delivering a constant 0.2 mA across a 6-10 V output voltage range.
- Modeled and analyzed buck and boost DC-DC converters, evaluating steady-state voltage conversion ratios and switching behavior.
- Designed and simulated a common-source MOSFET amplifier, achieving a voltage gain of 25 V/V at a 0.1 mA bias current.

**Successive Approximation ADC Implementation**

- Designed and implemented an SAR ADC prototype using discrete analog components.
- Developed the successive approximation logic using Arduino and interfaced it with the DAC and comparator.
- Integrated and debugged the complete ADC system, verifying analog-to-digital conversion.

**RISC-V Processor Design using Verilog**

- Implemented a single-cycle RISC-V processor in Verilog HDL supporting arithmetic, logical, memory, and branch instructions.
- Designed key modules including ALU, Register File, Control Unit, Immediate Generator, and Data Memory.
- Verified processor functionality through comprehensive Verilog testbenches. Designed and implemented a 32-bit RISC-V processor in Verilog HDL, progressing from single-cycle to a 5-stage pipelined architecture (IF, ID, EX, MEM, WB).
- Implemented data forwarding, hazard detection, and dynamic branch prediction using BHT and BTB with pipeline recovery.
- Verified the processor using Verilog testbenches, Icarus Verilog, and GTKWave.

**Battery Management System using ADBMS1818**

- Developed a battery monitoring system for multi-cell battery packs using the ADBMS1818 battery monitor IC.
- Implemented SPI communication with Arduino for real-time cell voltage acquisition.
- Performed PCB debugging, software validation, and troubleshooting of voltage measurement errors.

**Quantum Gate Leakage Current Simulation using Transfer Matrix Method (TMM)**

- Developed a numerical framework in C to simulate quantum tunneling and gate leakage current in MOS structures using the Transfer Matrix Method (TMM) and the Tsu-Esaki framework.
- Integrated electrostatic potential profiles from a self-consistent 1D nonlinear Poisson solver to compute energy-dependent transmission probabilities.
- Implemented the Tsu-Esaki leakage current model in C and used Python to visualize and validate leakage current simulations across multiple oxide barrier profiles.

## POSITIONS OF RESPONSIBILITY

**Transport Coordinator | Elan & nVision, IIT Hyderabad**

- Coordinated transportation logistics for participants and guests during IIT Hyderabad's annual technical and cultural fest.

**Core | Electronica, IIT Hyderabad**

- Designed and delivered a hands-on RISC-V workshop covering Verilog HDL and FPGA implementation for club members.